

PHASE 1 · 2012–2020

Disembodied ML

The Cloud Era

Systems Focus:

High-throughput serving of digital predictions
($x \rightarrow y$)

- **Substrate:** Cloud GPU Clusters
- **Workloads:** ResNet, BERT, RecSys
- **Boundary:** Stateless API / Screen
- **Failure Mode:** try/catch $\rightarrow$ Retry
- **Physical Action:** None (Digital bits)

Compress

PHASE 2 · 2018–2023

Edge Perception

The TinyML Era

Systems Focus:

Compressing models onto constrained microcontrollers

- **Substrate:** Bare-Metal MCUs / DSPs
- **Workloads:** Wake-words, Anomaly
- **Boundary:** Open-loop edge sensing
- **Failure Mode:** Dropped wake-up / Alert
- **Physical Action:** Passive telemetry

Reason

PHASE 3 · 2023–2026

Deliberation

The Foundation Era

Systems Focus:

Spatial reasoning using multimodal foundation models

- **Substrate:** Edge MPUs / NPUs
- **Workloads:** VLMs, Spatial Transformers
- **Boundary:** Semantic planning (1 Hz)
- **Failure Mode:** Hallucination, P_{99} tails
- **Physical Action:** Unverified proposals

Close Loop

PHASE 4 · NOW (THE FRONTIER)

Physical AI Systems

Closed-Loop Actuation

Systems Focus:

Learned proposals governed by real-time safety enforcers

- **Substrate:** Linux MPU + Real-Time MCU
- **Workloads:** Multi-Rate VLA + 1 kHz CBF
- **Boundary:** Delegated physical authority
- **Failure Mode:** Gearbox shear / Collisions
- **Physical Action:** Matter & kinetic energy

← Open-Loop & Digital Sandboxes (Software retries, idempotent computation, no motor coils)

Physical Causality ($W_t \rightarrow W_{t+1}$, No ctrl+z) →